$$w_p = C_{\tau,I_p}, \quad C_{\tau,0} = 0, \qquad \vartheta_\tau = (C_\tau, \theta_\tau)$$

SOURCE TASKS $\tau \in \mathcal{T}_{\mathrm{tr}}$

$I_{\tau r} \sim \nu_0$
or
$I_{\tau r} \leftarrow \mathrm{Extract}\big(A_{T_s}(S^{\mathrm{s}}_\tau; \xi_r)\big)$

$\widetilde{I} \leftarrow \mathrm{Align}(I)$
$\widehat{\vartheta} \leftarrow A_T(S^{\mathrm{s}}_\tau, \widetilde{I}; \xi')$
$\mathcal{B} = \{\widetilde{I} : \mathcal{L}_{S^{\mathrm{q}}_\tau}(f_{\widetilde{I},\widehat{\vartheta}}) \le \eta\}$

$Q_\phi(I) = \prod_{p=1}^{P} q_\phi(I_p \mid I_{<p})$
$\widehat{\phi} = \arg\min_\phi - \sum_{I \in \mathcal{B}} \log Q_\phi(I)$

$Q_{\widehat{\phi}}$

TARGET TASK $\tau \notin \mathcal{T}_{\mathrm{tr}}$

$I_b \sim Q_{\widehat{\phi}}, \qquad b = 1, \dots, B$
$\widehat{\phi}$ frozen, $\qquad I_b$ fixed

$\vartheta_b^{(0)} \sim \mu_0$
$\widehat{\vartheta}_b \leftarrow A_T(S^{\mathrm{s}}_\tau, I_b; \xi_b)$

$b^* = \arg\min_b \mathcal{L}_{S^{\mathrm{q}}_\tau}(f_{I_b,\widehat{\vartheta}_b})$
$\widehat{f}_\tau = f_{I_{b^*},\widehat{\vartheta}_{b^*}}$

$S^{\mathrm{s}}_\tau \perp S^{\mathrm{q}}_\tau \perp S^{\mathrm{test}}_\tau, \qquad \mathcal{L}_{S^{\mathrm{test}}_\tau}(\widehat{f}_\tau)$